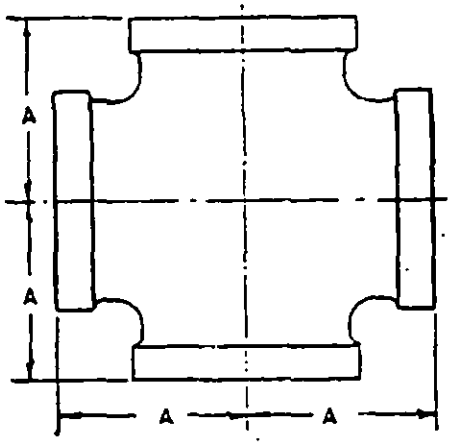
MILITARY SPECIFICATION SHEET

FITTINGS, PIPE, CAST BRONZE, SILVER-BRAZING; CROSS, PIPE

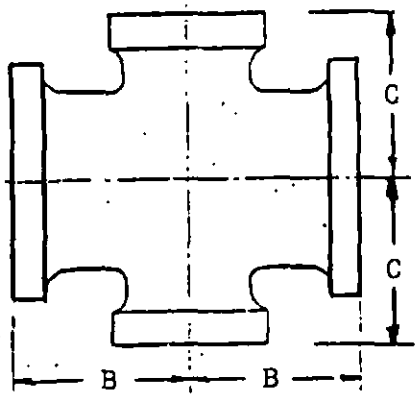
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crosses described herein shall consist of this document and the latest issue of specification MIL-F-1183.



SH 12162

FIGURE 1. Cross.



SH 12396

FIGURE 2. Reducing cross.

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TABLE I. Center-to-end dimensions.

Nominal pipe size (Inches)	Center-to-end (Inches)		
	A	B	C
1/4 (all ends)	0.71		
3/8 x 1/4		0.75	0.78
3/8 (all ends)	.82		
1/2 x 3/8 x 3/8 x 3/8		.81	.90
1/2 x 3/8		.93	.90
1/2 (all ends)	1.01		
3/4 x 3/8		1.00	1.00
3/4 x 1/2		1.08	1.11
3/4 (all ends)	1.18		
1 x 3/8		1.20	1.24
1 x 1/2		1.20	1.24
1 x 3/4		1.30	1.31
1 (all ends)	1.43		
1-1/4 x 1/2		1.29	1.41
1/ 1-1/4 x 3/4		1.52	1.60
1-1/4 x 1		1.52	1.60
1-1/4 (all ends)	1.69		
1-1/2 x 3/4		1.42	1.60
1-1/2 x 1		1.55	1.72
1-1/2 x 1-1/4		1.72	1.81
1-1/2 (all ends)	1.84		
2 x 3/4		1.47	1.84
2 x 1		1.59	1.95
2/ 2 x 1-1/4		1.89	2.07
2 x 1-1/2		1.89	2.07
2 (all ends)	2.12		
2-1/2 x 2		2.39	2.60
2-1/2 (all ends)	2.70		
3 x 2-1/2		2.83	2.99
3 (all ends)	3.08		
3-1/2 (all ends)	3.42		
4 x 3		3.30	3.60
4 (all ends)	3.79		
5 x 4		4.00	4.41
5 (all ends)	4.50		
6 (all ends)	5.13		
8 (all ends)	6.56		

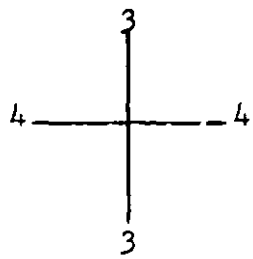
1/ Bushed from 1-1/4 x 1.
2/ Bushed from 2 x 1-1/2.

1. *Cross type*. Regular crosses (see figure 1) shall be type A (streamlined interior).

2. Reducing cross type. Reducing crosses (see figure 2) shall be type B (interior not streamlined).

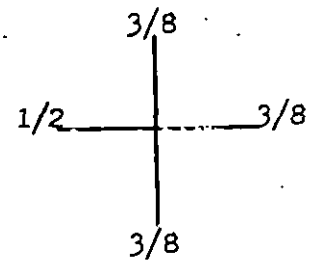
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3. Identification of reducing cross end sizes. In identifying reducing cross end sizes, the run ends are given first and then the outlet ends are given last. A cross with reducing outlets as shown on figure 3 is identified as a 4 x 3 inch nominal pipe size reducing cross. A cross with reduced run and outlets as specified on figure 4 is identified as a 1/2 x 3/8 x 3/8 x 3/8 inch nominal pipe size reducing cross.



SH 12163

FIGURE 3. Reduced outlet.



SH 12164

FIGURE 4. Reduced run and outlet.

4. Center-to-end dimensions. The center-to-end dimensions for crosses and reducing crosses shall be as specified in table I.

5. Military specification sheet part number. Crosses covered by this specification sheet shall be designated in the following form (see 6.2(b) of MIL-F-1183).

M1183/4 - XX X

Military specification sheet code number
Cross style and pipe size code number (see requirement 6 herein)
Brazing ring groove and brazing ring code letter (see requirement 7 herein)

6. Cross style and pipe size. The cross style (cross or reducing cross) and nominal pipe sizes(s) are identified by two digits (see table II).

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TABLE II. Code number to style and size.

Cross (type A)	Reducing cross (type B)	Nominal pipe size (Inches)
01		1/4 (all ends)
	21	3/8 x 1/4
02		3/8 (all ends)
	22	1/2 x 3/8 x 3/8 x 3/8
	23	1/2 x 3/8
03		1/2 (all ends)
	24	3/4 x 3/8
	25	3/4 x 1/2
04		3/4 (all ends)
	26	1 x 3/8
	27	1 x 1/2
	28	1 x 3/4
05		1 (all ends)
	29	1-1/4 x 1/2
	30	1-1/4 x 3/4
	31	1-1/4 x 1
06		1-1/4 (all ends)
	32	1-1/2 x 3/4
	33	1-1/2 x 1
	34	1-1/2 x 1-1/4
07		1-1/2 (all ends)
	35	2 x 3/4
	36	2 x 1
	37	2 x 1-1/4
	38	2 x 1-1/2
08		2 (all ends)
	39	2-1/2 x 2
09		2-1/2 (all ends)
	40	3 x 2-1/2
10		3 (all ends)
11		3-1/2 (all ends)
	41	4 x 3
12		4 (all ends)
	42	5 x 4
13		5 (all ends)
14		6 (all ends)
15		8 (all ends)

7. Brazing ring groove and brazing ring. Face fed (no groove), grooved without brazing ring, grooved with brazing ring inserted in groove, grooved with brazing ring(s) separately packaged and attached to the cross or grooved with brazing ring(s) separately packaged and shipped with but not attached to the cross socket ends are identified by a single letter (see table III).

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TABLE III. Code letter to socket condition.

Code letter	Condition of socket
N	No groove (face fed)
G 1/	Grooved (no brazing ring)
R 1/	Grooved with brazing ring inserted
A 2/	Grooved with brazing ring separately packaged and attached
S 1/	Grooved with brazing ring separately packaged and shipped with

1/ No National Stock Number (NSN) shall be established or exist in the Federal Supply System for cross with the code letter of G, R or S.

2/ When the socket condition code letter is A, the brazing rings are not to be inserted in the grooves but packaged separately and either included within each individual cross package or physically attached to the cross.

8. Military specification part number definition. The military specification sheet part number is a definitive part number which is formed by combining the military specification sheet code number (M1183/4 for MIL-F-1183/4) with a dash after it, the cross style and pipe size code number and the brazing ring groove and brazing ring code letter as specified in requirement 5.

Custodians:
Army - ME
Navy - SH
Air Force - 99

Review activity:
DLA - CS

Preparing activity:
Navy - SH

Agent:
DLA - CS

(Project 4730-0047)

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